ALU Construction

ALU (Arithmetic Logic Unit)

- arithmetic ops.
- logical ops.
- branch computation
- address computation

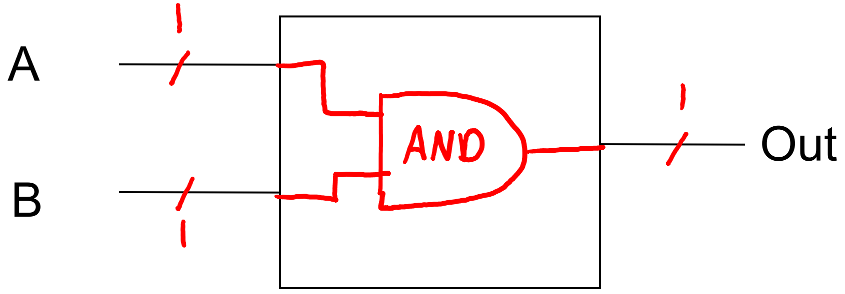
Decide which operations to support

Operation	OpCode
1. And	00
2. Or	01
3. Add	10
4. Sub	11

- Choose operations
- Assign op-codes

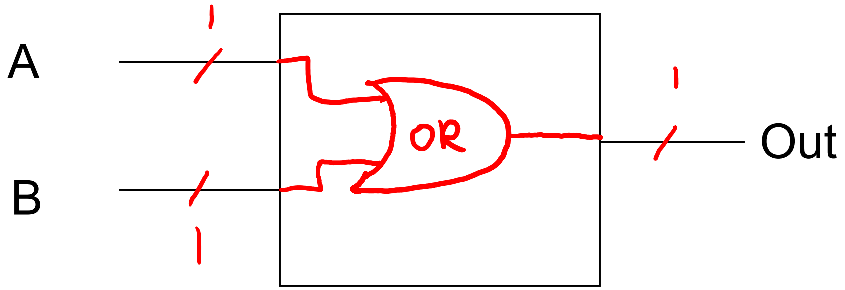
Design 1-bit for each operation And & Or

1.



A & B

2.



A | B

Design 1-bit for each operation

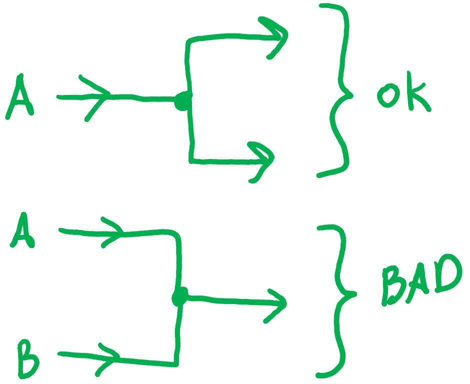
3. Add

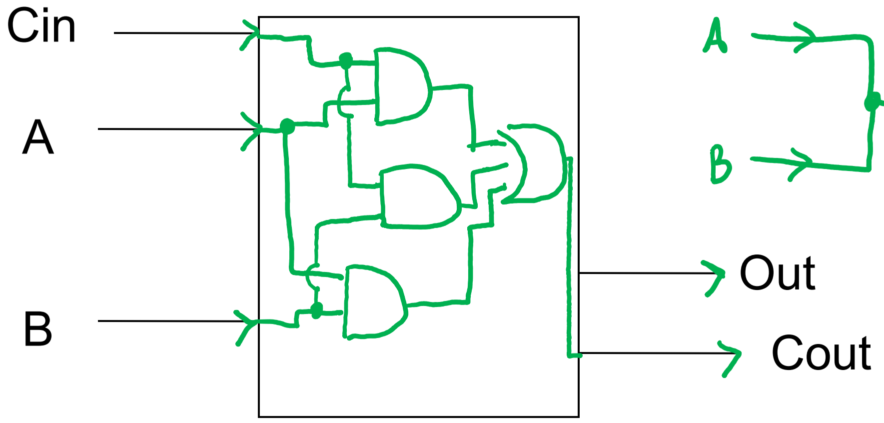
$$\underbrace{A, B, C_{in}}_{\text{inputs}} \qquad \underbrace{C_{out}, Out}_{\text{outputs}}$$

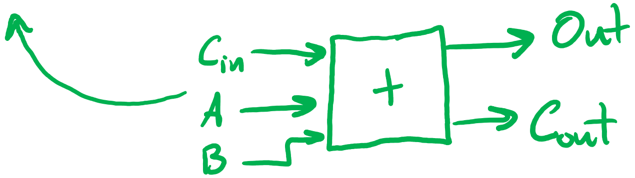
A	B	Cin	Cout	Out
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

$$C_{out} = A \overset{\text{and}}{\cdot} B \overset{\text{or}}{+} A \cdot C_{in} + B \cdot C_{in}$$

$$Out = A \text{ XOR } B \text{ XOR } C_{in}$$







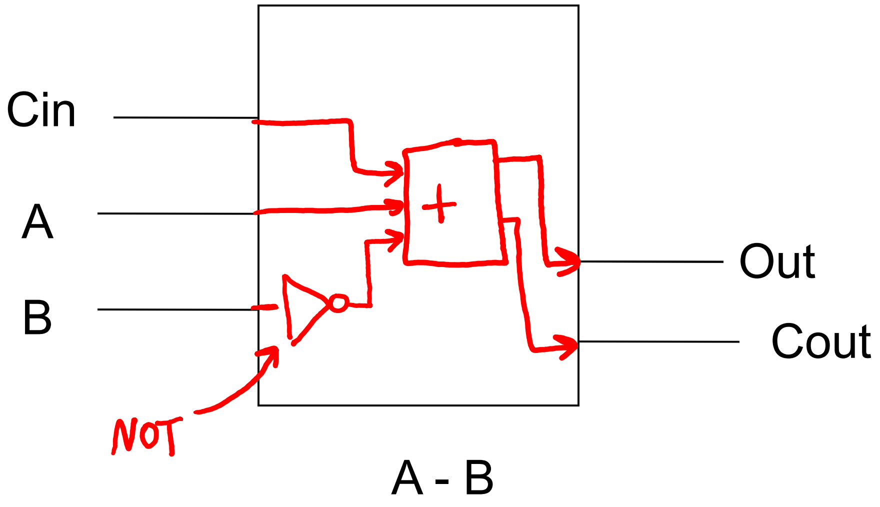
A + B

Cout
1 1 Cin

```
  0110   A
+ 0011   B
  ----
  1001   Out
```

Design 1-bit for each operation

4. Sub

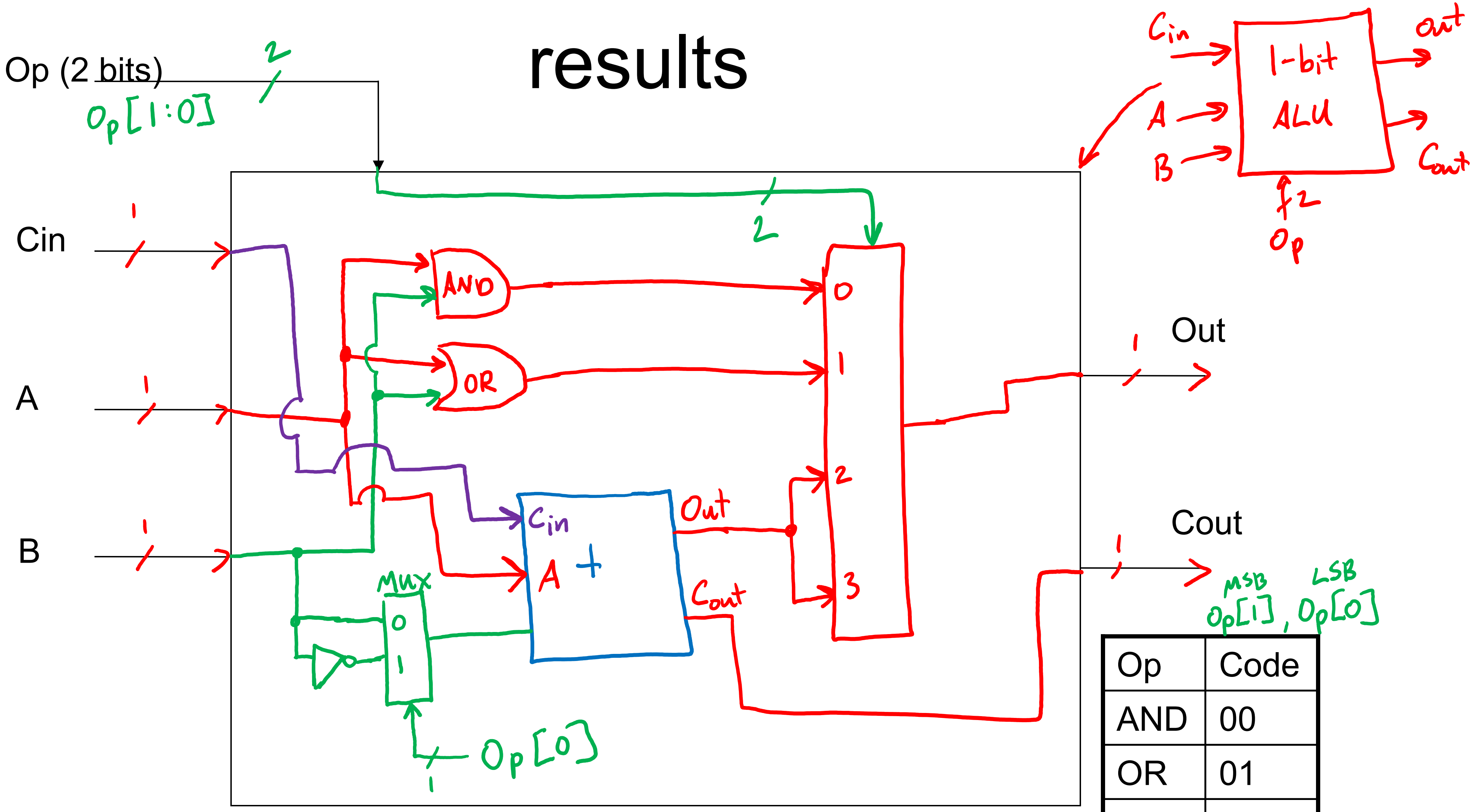


$A - B = A + (-B)$

- to compute subtract, flip B, but we need to add 1 later

$$\begin{array}{r} 0110 \\ -\ 0011 \rightarrow \\ \hline \end{array}$$

Use MUX to choose between results



Op	Code
AND	00
OR	01
Add	10
Sub	11

Design logic to connect ALU